

Name: _____

Incoming Statistics Graduate Students Computer Programming Proficiency Test

The purpose of this placement test is to determine whether you have already met the programming proficiency requirement in our statistics graduate program. If you do not have previous experience or coursework in programming, you may opt not to take this test. Students who are not deemed proficient will be required to do an extra programming practice assignment in STAT:5400.

For the following programming problems, you may use your choice of the following programming languages:

- R (preferred)
- Python
- C
- C++
- Fortran

Print your answers on a separate sheet of paper; be sure to put your name on it.

No submission is required, but ideally you can try this before Friday's class.

1. Write a function or subroutine that will solve quadratic equations.

Recall that a quadratic equation has the form

$$ax^2 + bx + c = 0$$

If the equation has real roots, they can be computed from the formulas:

$$\begin{aligned}x_1 &= \frac{-b + \sqrt{b^2 - 4ac}}{2a} \\x_2 &= \frac{-b - \sqrt{b^2 - 4ac}}{2a}\end{aligned}$$

If the quantity $b^2 - 4ac$ (called the *discriminant*) is negative, the equation will not have real roots.

Your function should

- (a) accept as its arguments the coefficients a , b , and c .
- (b) compute the discriminant

- (c) return three numbers as indicated below (the third number is an indicator as to whether real roots exist)
- if the discriminant is non-negative,
compute the two roots;
return the two roots and a 1
 - if the discriminant is negative,
return two values of -99 and a 0

If your language of choice permits you to set the precision of numbers, all arguments and returned values should be in double precision.

2. Write a main calling program that does the following:
- (a) Initializes a double-precision array with 10 rows and 6 columns.
 - (b) Assigns the following values to the first 3 columns of the array:
 - i. column 1: all 1's
 - ii. column 2: the row number
 - iii. column 3: the square root of the row number
 - (c) For each row of the array:
 - i. Calls your function from question 1, passing the entries in the first 3 columns of the array as the arguments.
 - ii. Assigns the values returned by the function to the last three columns in the row.